STA-100

General math & Bio stat

| Mid term |
FOr MSC Zoology they are for
Final term syllabus

Designed & Composed by:

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ب ٹ

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Question No: 1 (Marks: 1) - Please choose one

10! =…….

► 362880

*► 3628800

► 362280

► 362800

Question No: 2 (Marks: 1) - Please choose one

If a player well shuffles the pack of 52 playing card, then the probability of a black card from 52 playing cards is: http://vustudents.ning.com

► $\frac{1}{52}$

*► $\frac{13}{52}$

► $\frac{4}{52}$

► $\frac{26}{52}$

Question No: 3 (Marks: 1) - Please choose one

The probability of drawing a 'jack card ' from 52 playing cards is:

► $\frac{1}{52}$

*► $\frac{4}{52}$

► $\frac{13}{52}$

► $\frac{26}{52}$

Question No: 4 (Marks: 1) - Please choose one

Which dispersion is used to compare variation of two series:

► C.V.

*► Q.D.

► M.D.

► S.D.

Question No: 5 (Marks: 1) - Please choose one

If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r': http://vustudents.ning.com

► $0 \leq r \leq 1$

► $r \geq 0$

*► $r = +1$

► r=-1

Question No: 6 (Marks: 1) - Please choose one

In a regression line Y= a + bX, the value of the correlation coefficient will be zero if:

*► Intercept a = 0

► Intercept $a \neq 0$

► Slope b = 0

► Slope $b \neq 0$

Question No: 7 (Marks: 1) - Please choose one

When two coins are tossed the probability of at least one head is:

► 1/4

* ► 3/4

► 2/4

► 4/4

Question No: 8 (Marks: 1) - Please choose one

Which one of the following measurement does *not* divide a set of observations into equal parts?

► quartiles

► deciles
► percentiles
*► standard deviations

Question No: 9 (Marks: 1) - Please choose one

In the model Y = mX + a, Y is also known as the:

► Predictor variable
► Independent variable
► Predicted variable
► Explanatory variable

Question No: 10 (Marks: 1) - Please choose one

According to empirical rule approximately 95% of the measurements will fall under which interval? http://vustudents.ning.com

► $\bar{X} \pm S$
*► $\bar{X} \pm 2S$
► $\bar{X} \pm 3S$
► $\bar{X} \pm 4S$

Question No: 11 (Marks: 1) - Please choose one

Which one of the following is written at the top of the table?

► Source note
► Foot note
► Prefatory note
*► Title

Question No: 12 (Marks: 1) - Please choose one

If a curve has a longer tail to the right, it is called :

*► Positively skewed
► Negatively skewed
► J-shaped

► Symmetric

Question No: 13 (Marks: 1) - Please choose one

Which one of the following is the class frequency?

* ► The number of observations in each class
► The difference between consecutive lower class limits
► Always contains at least 5 observations
► Usually a multiple of the lower limit of the first class

Question No: 14 (Marks: 1) - Please choose one

If X is a discrete random variable, then the function $f(x)$ is

► A probability function
► A probability density function
► A density function
*► A distribution function

Question No: 15 (Marks: 1) - Please choose one

Which one of the following graphs is used for a time series data?

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*► Histogram

► Historigram
► Frequency curve

► Frequency polygon

Question No: 16 (Marks: 1) - Please choose one

If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

► Ogive
► Frequency polygon
*► Frequency curve
► Historigram

Question No: 17 (Marks: 1) - Please choose one

If mean of the two observations is 10.5, then median of these two observations will be:

► 7.5
► 8.5
► 9.5
*► 10.5

Question No: 18 (Marks: 1) - Please choose one

Which one is the formula of mid range:

► $x_m - x_0$
► $x_0 - x_m$
► $\frac{x_0 - x_m}{2}$
*► $\frac{x_0 + x_m}{2}$

Question No: 19 (Marks: 1) - Please choose one

Which one of the following is not included in measures of central tendency:

*► Quartile deviation

► Harmonic mean

► Geometric mean

► Arithmetic mean

Question No: 20 (Marks: 1) - Please choose one

For the given data 2, 3, 7, 0, -8 G. M will be:

► Negative
► Positive
► Zero
* ► Undefined

Question No: 21 (Marks: 2)

Why measure of central tendency and measure of dispersion are complementary to each other?

Question No: 22 (Marks: 2)

What do you know about discrete random variable?

Ans:

Such a numerical quantity whose value is determined by the value of a random experiment is called a random variable.

Example:

If we toss three dice together and let X represents the number of heads, then the random variable X consists of the value 0, 1, 2, and 3. the X in this example is a discrete random variable.

Question No: 23 (Marks: 3)

What is the subjective approach to the probability?

Ans:

Subjective probability is a measure of the strength of a person's belief regarding the occurrence of an event A. Probability in this sense is purely subjective and is based on whatever evidence is available to the individual.

It has a disadvantage that 2 or more persons faced with the same evidence may arrive at different probabilities.

Example:

Suppose a panel of three judges is hearing a trial. It is possible that based on the evidence that is presented, two of them arrive at the conclusion that the accused is guilty while one of them decides that the evidence is not strong to draw this conclusion.

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Question No: 24 (Marks: 3)

Explain the difference between absolute dispersion and relative dispersion:

Ans:

Question No: 25 (Marks: 5)

Differentiate between the mutually exclusive events and exhaustive events.

Mutually Exclusive Events
Two events A and B of a single experiment are said to be mutually exclusive if and only if they cant both occur at the same time.

Example:

When a die is rolled, the events ‘even number’ and odd number are mutually exclusive as we can get eithere an even number or an odd number in one throw, not both at the same time.

Exhaustive Events:

Events are said to be collectively exhaustive, when the union of mutually exclusive events is equal to the entire sample space S.

Examples

In the coin tossing experiment, head and tail are collectively exhaustive events.

Question No: 26 (Marks: 5)

Find the first two moments about mean from the following data.
X= 34, 70, 42, 54, 40, 68, 56, 38, 36, 72

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MIDTERM EXAMINATION

Spring 2009

STA301- Statistics and Probability (Session - 6)

Time: 60 min

Marks: 38

Question No: 1 (Marks: 1) - Please choose one

For a positively skewed distribution m_3 will be:

► Positive

► Negative

► Zero

► 1

Question No: 2 (Marks: 1) - Please choose one

When data is labeled to identify an attribute of element, the measurement scale is:

► Ordinal

► Interval

► Nominal

► Ratio

Question No: 3 (Marks: 1) - Please choose one

Suppose the estimated equation is has been calculated for a set of data. What is slop of the line:

► 0

► 2

► -2

► 5

Question No: 4 (Marks: 1) - Please choose one

If P(B|A) = 0.25 and , then P(A) is:

► 0.05

► 0.80

► 0.95

► 0.75

Question No: 5 (Marks: 1) - Please choose one

Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:

► Advance statistics

► Probability statistics

► Descriptive statistics

► Inferential statistics

Question No: 6 (Marks: 1) - Please choose one

In a sample of 800 students in a university, 160, or 20%, are Business majors. Based on the above information, the school's paper reported that "20% of all the students at the university are Business majors." This report is an example of :

► A sample

► A population

► Sstatistical inference

► Descriptive statistics

Question No: 7 (Marks: 1) - Please choose one

A set that contains all possible outcomes of a system is known as

► Finite Set

► Infinite Set

► Universal Set

► No of these

Question No: 8 (Marks: 1) - Please choose one

If X and Y are independent, then Var(X-Y) is equal to:

►

►

►

► Zero

Question No: 9 (Marks: 1) - Please choose one

Which of the following is the class frequency

► The number of observations in each class

► The difference between consecutive lower class limits

► Always contains at least 5 observations

► Usually a multiple of the lower limit of the first class

Question No: 10 (Marks: 1) - Please choose one

How to construct the class interval:

► Divide the class frequencies in half

► Divide the class frequency by the number of observations

► Find the difference between consecutive lower class limits

► Count the number of observations in the class

Question No: 11 (Marks: 1) - Please choose one

Data in the Population Census Report is:

► Ungrouped data

► Secondary data

► Primary data

► Arrayed data

Question No: 12 (Marks: 1) - Please choose one

What is the range of -2,-3,-5,-10 :

► -12

► 8

► -8

► 2

Question No: 13 (Marks: 1) - Please choose one

The algebraic sum of deviations from mean is:

► Maximum

► Minimum

► Zero

► Undefined

Question No: 14 (Marks: 1) - Please choose one

The sum of squares of deviations from mean is:

► Undefined

► Zero

► Maximum

► Minimum

Question No: 15 (Marks: 1) - Please choose one

Statistic is a numerical quantity, which is calculated from:

► Population

► Sample

► Data

► Observations

Question No: 16 (Marks: 1) - Please choose one

Which of the following is not based on all the observations?

► Arithmetic Mean

► Geometric Mean

► Harmonic mean

► Mode

Question No: 17 (Marks: 1)

Elaborate the word dispersion.

We can say that the degree of scatter of data, usually about an average value, can be the median.

Question No: 18 (Marks: 1)

Define population.

We can define population is the collection of individuals or objects having some common measurable characteristics.

Question No: 19 (Marks: 2)

What does mean by the independence of two events:

We can define independence of two events are statistically independent if the probability of their occurring jointly equals the product of their respective probabilities. Independence of two events also know as stochastic independence.

Question No: 20 (Marks: 3)

The reciprocal of the values are

0.012, 0.0235, 0.0135

Calculate Harmonic Mean

Harmonic mean is 250.037

Question No: 21 (Marks: 5)

The probability that a student passes mathematics is 2/3 and the probability that he passes English is 4/9. If the probability of passing at least one course is 4/5, what is the probability that he will pass both courses?

Math=2/3

English=4/9

Least one passing probability= 4/5

Math + English =2/3+4/9

=1.11

4/5+1.11

=1.911

Question No: 22 (Marks: 10)

A pair of dice is thrown, then

1) Find the sample space for this experiment

Suppose if A and B is the pair of dice then lets try to find out how many result we can find

- A hit first
- B Hit first
- B hit Second
- A Hit Second
- Both Hit Equally
- Both didn't Hit

2) Determine the probability of getting the sum 8 on the dice

3) Find the probability of getting sum 7 or 11

Question # 1 of 10 (Start time: 12:11:21 AM) Total Marks: 1

Let X be a random variable with binomial distribution, that is (X=0,1,..., n). The expected value E[X] is

Select correct option:

- p
- np
- np(1-p)
- Xnp

Question # 2 of 10 (Start time: 12:12:17 AM) Total Marks: 1

The sample mean is an unbiased estimator for the population mean. This means:

Select correct option:

- The sample mean has a normal distribution
- The average sample mean, over all possible samples, equals the population mean
- The sample mean is always very close to the population mean
- The sample mean will only vary a little from the population mean

Question # 3 of 10 (Start time: 12:13:17 AM) Total Marks: 1

Probability of an impossible event is always:

Select correct option:

- Less than one
- Greater than one
- Between one and zero
- Zero

Question # 4 of 10 (Start time: 12:13:48 AM) Total Marks: 1

The function abbreviated to d.f. is also called the.....

Select correct option:

- Probability density function
- Probability distribution function
- Commutative distribution function
- Discrete function

Question # 5 of 10 (Start time: 12:14:50 AM) Total Marks: 1

The total area under the normal curve is:

Select correct option:

- 0
- 1
- 0.5
- 0.75

Question # 6 of 10 (Start time: 12:15:12 AM) Total Marks: 1

Two events A & B are said to be independent if....

Select correct option:

- P (A) + P (B)
- P (B\A) = P (B)
- P (A) * P (B)

P (A\B) = P (A)

Question # 7 of 10 (Start time: 12:15:31 AM) Total Marks: 1

When two coins are tossed the probability of at most one head is:

Select correct option:

1/4

2/4

3/4

1

Question # 8 of 10 (Start time: 12:16:33 AM) Total Marks: 1

For exhaustive events, the P(AUBUC) is equal to:

Select correct option:

P(A)

P(S)

P(A) * P(B)* P(C)

P(B)

Question # 9 of 10 (Start time: 12:17:46 AM) Total Marks: 1

One card is drawn from a standard 52 card deck. In describing the occurrence of two possible events, an Ace and a King, these two events are said to be:

Select correct option:

independent

randomly independent

random variables

mutually exclusive

Question # 10 of 10 (Start time: 12:18:23 AM) Total Marks: 1

The number of parameters in hypergeometric distribution is (are):

Select correct option:

1

2

3

4

Sta301
Select the most suitable option.
1. Which is not a source of existing statistics?
A. Government
B. International agencies
C. Personal interviews
D. Private sources
2**. Followings are the advantages of secondary data except**
A. Non-Reactivity
B. Selective survival
C. Low cost
D. Spontaneity
3. Following are the disadvantages of secondary data except
A. Longitudinal analysis
B. Sampling bias
C. Coding difficult
D. Incompleteness
4. The coding of the secondary data may be difficult because of
A. Differences in content or subject matter
B. Lack of standardization
C. Differences in length and format
D. All of these
5. The researcher must be concerned about the following problems while using secondary data in research.
A. Validity
B. Reliability
C. Both of these
D. None of these

Research Methods (STA630)
Fall Semester 2006
(Solution) Quiz # 01
Marks 10

6- Hypothesis refers to

A. The outcome of an experiment

B. A conclusion drawn from an experiment

C. A form of bias in which the subject tries to outguess the experimenter

D. A tentative statement about the relationship

7- Statistics is used by researchers to

A. Analyze the empirical data collected in a study

B. Make their findings sound better

C. Operationally define their variables

D. Ensure the study comes out the way it was intended

8- A literature review requires

A. Planning

B. Good & clear writing

C. Lot of rewriting

D. All of the above

9- A literature review is based on the assumption that

A. Copy from the work of others

B. Knowledge accumulates and learns from the work of others

C. Knowledge disaccumulates

D. None of the above option

10- A theoretical framework

A. Elaborates the r/s among the variables

B. Explains the logic underlying these r/s

C. Describes the nature and direction of the r/s

D. All of the above

11- Which of the following statement is not true?

A. A research proposal is a document that presents a plan for a project

B. A research proposal shows that the researcher is capable of successfully conducting the proposed research project

C. A research proposal is an unorganized and unplanned project

D. A research proposal is just like a research report and written before the research project

12- Preliminary data collection is a part of the

A. Descriptive research

B. Exploratory research

C. Applied research

D. Explanatory research

13- Conducting surveys is the most common method of generating

A. Primary data

B. Secondary data

C. Qualitative data

D. None of the above

14- After identifying the important variables and establishing the logical reasoning in theoretical framework, the next step in the research process is

A. To conduct surveys

B. To generate the hypothesis

C. To focus group discussions

D. To use experiments in an investigation

15- The appropriate analytical technique is determined by

A. The research design

B. Nature of the data collected

C. Nature of the hypothesis

D. Both A & B

16-Personal interviews conducted in shopping malls are known as:

A. Mall interviews
B. Mall intercept interviews
C. Brief interviews
D. None of the given options

17-WATS lines provided by long distance telephone service at fixed rates. In this regard, WATS is the abbreviation of:

E. West Africa Theological Seminary
F. Washtenaw Area Transportation Study
G. Wide Area Telecommunications Service
H. World Air Transport Statistics

18-A list of questions which is handed over to the respondent, who reads the questions and records the answers himself is known as the:

I. Interview schedule
J. Questionnaire
K. Interview guide
L. All of the given options

19-One of the most critical stages in the survey research process is:

M. Research design
N. Questionnaire design
O. Interview design
P. Survey design

20-Question that consists of two or more questions joined together is called a:

Q. Double barreled question
R. General question
S. Accurate question
T. Confusing question

21-The number of questionnaires returned or completed divided by the total number of eligible people who were contacted or asked to participate in the survey is called the:

U. Response rate
V. Participation rate

W. Inflation rate
X. None of the given options

22-To obtain the freest opinion of the respondent, when we ask general question before a specific question then this procedure is called as the:

Y. Research technique
Z. Qualitative technique
AA. Funnel technique
BB. Quantitative technique

23-A small scale trial run of a particular component is known as:

CC. Pilot testing
DD. Pre-testing
EE. Lab experiments
FF.Both A & B

24-Field testing of the questionnaire shows that:

GG. Respondents are willing to co-operate
HH. Respondents are not willing to co-operate
II. Respondents do not like any participation
JJ. All of the given options

25- Service evaluation of hotels and restaurants can be done by the:

KK. Self-administered questionnaires
LL. Office assistant
MM. Manager
NN. None of the given options

26. _____research is based on naturalism.
A. Field research
B. Descriptive research
C. Basic research
D. Applied research

27- Personal interviews conducted in shopping malls are known as_____
A. Mall interviews
B. Mall intercept interviews
C. Brief interviews
D. None of the given options

28. _____is used to obtain the freest opinion of the respondent, by asking general question before a specific question.
A. Research technique
B. Qualitative technique
C. Funnel technique

D. Quantitative technique

29. In, _____the interviewer and members jointly control the pace and direction of the interview.

A. Field interview

B. Telephonic interview

C. Both A and B

D. None of the given options

30. Randomization of test units is a part of _____

A. Pretest

B. Posttest

C. Matching

D. Experiment

31. Which one of the following sets is the measure of central tendency?

a. Mean, standard deviation, mode

b. Mean, median, standard deviation

c. Arithmetic mean, median, mode

d. Standard deviation, internal validity, mode

32. Internal validity refers to .

a. Researcher's degree of confidence.

b. Generalizability

c. Operationalization

d. All of the above

33. How many times the students appear in the research class is the example of _____.

a. Intensity

b. Space

c. Frequency

d. Direction

34. Time consumed in mall intercept interview is .

a. High

b. Moderate

c. Low

d. Nil

35. Departmental stores selected to test a new merchandising display system is the example of .

a. Quota sampling

b. Convenience sampling

c. Judgmental sampling

d. Purposive sampling

36. In _____, the researcher attempts to control and/ or manipulate the variables in
the study.

1. Experiment

2. Hypothesis

3. Theoretical framework

4. Research design

36. In an experimental research study, the primary goal is to isolate and identify the effect produced by the ____.
1. Dependent variable
2. Extraneous variable
37. Independent variable
4. Confounding variable
3. A measure is reliable if it provides consistent _____.
1. Hypothesis
2. Results
3. Procedure
4. Sensitivity
38. The interview in which questions are already prepared is called _____.
1. Telephonic interview
2. Personal interview
3. Unstructured interview
4. Structured interview
39. The numerical description that describe sample may be expected to differ from those that describe population because of random fluctuations inherent in sampling process.
1. Sampling design
2. Non-probability sampling
3. Sampling error
4. Probability sampling
40. In _____ , each population element has a known and equal chance of selection.
1. Purposive sampling
2. Quota sampling
3. Stratified sampling
4. Simple random sampling
41. _____ is the evidence that the instrument, techniques, or process used to measure concept does indeed measure the intended concepts.
1. Reliability
2. Replicability
3. Scaling
4. Validity
42. A researcher is interested in studying why the "new math" of the 1960s failed. She interviews several teachers who used the new math during the 1960s. These teachers are considered as:
1. Primary sources
2. Secondary Sources
3. External critics
4. Internal critics
42. Which of the following is NOT true about stratified random sampling?
1. It involves a random selection process from identified subgroups
2. Proportions of groups in the sample must always match their population proportions
3. Disproportional stratified random sampling is especially helpful for getting large

enough subgroup samples when subgroup comparisons are to be done
4. Proportional stratified random sampling yields a representative sample
43. Experimental design is the only appropriate design where_____ relationship can be established.
1. Strong
2. Linear
3. Weak
4. Cause and Effect
45. All the persons involved in the collection of data and supervision of data collection process are called
i. Fieldworkers
ii. Researchers
iii. Research assistants
iv. None of the given options
46. While terminating the interview, the fieldworker should **not** do one of the following:
i. He should record all the responses made by the interviewee before leaving.
ii. He should thank the interviewee.
iii. He should close the interview hastily.
iv. He should answer all the questions the respondent asks concerning the nature and purpose of the study.
47. Which one of these is a type of Interviewee bias?
i. The respondent does not tell his true income, age, or contact information.
ii. The fieldworker fails to probe the interviewee properly.
iii. The fieldworker contacted the wrong person for interview.
iv. The fieldworker asks the questions in wrong order.
48. A magazine conducts a survey and asks its readers to cut the questionnaire from the magazine, fill it and send it via mail. It is a type of
i. Purposive sampling
ii. Snowball sampling
iii. Sequential sampling
iv. Convenience sampling
49. The height distribution of a few students in a school is an example of
i. Statistic
ii. Population
iii. Parameter
iv. Element
50. A researcher wants to conduct a survey of the drug users. Which type of sampling technique will be most appropriate here?
i. Sequential sampling
ii. Snowball sampling
iii. Quota sampling
iv. Convenience sampling
51. When there is a need to apply different data collection methods to different parts of the population, the best sampling method would be
i. Double sampling
ii. Cluster sampling

iii. Stratified random sampling
iv. Systematic random sampling
52. The sampling technique in which every element of the population has an equal, nonzero
probability of being selected in a sample, is called
i. Probability sampling
ii. Convenience sampling
iii. Purposive sampling
iv. Quota sampling
53. Target population is also called
i. Population
ii. Survey population
iii. Population element
iv. Population frame
54. Which one of them is the method for probing the respondent?
i. Repeat the question
ii. Give an expectant pause
iii. Repeat the respondent's reply
iv. All of the given options

Note: The correct options are highlighted in blue.

55. Which one of the following sets is the measure of central tendency?

a. Mean, standard deviation, mode
b. Mean, median, standard deviation
c. Arithmetic mean, median, mode
d. Standard deviation, internal validity, mode

56-In lab experiment the effect of _____Variables is controlled to evaluate the causal relationship.

a. Extraneous
b. Moderate
c. Intervening
d. All of the above

57-Internal validity refers to_____.

a. Researcher's degree of confidence.
b. Generalizability
c. Operationalization
d. All of the above

58-Which of the following is the weakest experimental design?

a. One group pretest-posttest design
b. Quasi- experimental design
c. Two group posttest only design
d. Ex post facto design

59-How many times the students appear in the research class is the example of _____.

a. Intensity
b. Space
c. Frequency
d. Direction

60-Disadvantage of content analysis is_____.

a. Researcher can increase the sample size
b. Provides access on the subjects to which researcher does have physical access.
c. Sometime documents provide incomplete account to the researcher
d. Spontaneous feelings can be recorded when they occurred

61-. Which of the following statement is incorrect with respect to "An experimental design is a set of procedures specifying:"

a. How the test units (subjects) are to be divided into homogenous sub samples.
b. What independent variables or treatments are to be measured?
c. What dependent variables are to be measured?
d. How the extraneous variables are to be controlled?

62-. Time consumed in mall intercept interview is _____.

a. High
b. Moderate
c. Low
d. Nil

63-. "Teacher should create a friendly environment in the classroom" this is the type of _____.

a. Leading question
b. Loaded question
c. Double Barreled
d. Burdensome question

64-. Departmental stores selected to test a new merchandising display system is the example of _____.

a. Quota sampling
b. Convenience sampling
c. Judgmental sampling
d. Purposive sampling

65-Discrete variable is also called…….

A. Categorical variable
B. Discontinuous variable
C. **Both A & B**
D. None of the above

66"Officers in my organization have higher than average level of commitment" Such a hypothesis is an example of…….

A. **Descriptive Hypothesis**
B. Directional Hypothesis
C. Relational Hypothesis
D. All of the above

67-'Science' refers to…….

A. A system for producing knowledge
B. The knowledge produced by a system
C. **Both A & B**
D. None of the above

68-Which one of the following is not a characteristic of scientific method?

A. Deterministic
B. Rationalism
C. Empirical
D. **Abstraction**

69-The theoretical framework discusses the interrelationships among the…….

A. **Variables**
B. Hypothesis
C. Concept
D. Theory

70-……research is based on naturalism.

A. **Field research**
B. Descriptive research
C. Basic research
D. Applied research

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O. Both A and B
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Q. Pretest
R. Posttest
S. Matching
T. **Experiment**

75. _____research is based on naturalism.
A. Field research
B. Descriptive research
C. Basic research
D. Applied research
76. Personal interviews conducted in shopping malls are known as_____
A. Mall interviews
B. Mall intercept interviews
C. Brief interviews
D. None of the given options
77. _____is used to obtain the freest opinion of the respondent, by asking general question before a specific question.
A. Research technique
B. Qualitative technique
C. Funnel technique
D. Quantitative technique
78. In, _____the interviewer and members jointly control the pace and direction of the interview.
A. Field interview
B. Telephonic interview
C. Both A and B
D. None of the given options
79. Randomization of test units is a part of _____
A. Pretest
B. Posttest
C. Matching
D. Experiment
80. Which one of the following sets is the measure of central tendency?
a. Mean, standard deviation, mode

b. Mean, median, standard deviation
c. Arithmetic mean, median, mode
d. Standard deviation, internal validity, mode
81. Internal validity refers to .
a. Researcher's degree of confidence.
b. Generalizability
c. Operationalization
d. All of the above
82. How many times the students appear in the research class is the example of _____.
a. Intensity
b. Space
c. Frequency
d. Direction
83. Time consumed in mall intercept interview is .
a. High
b. Moderate
c. Low
d. Nil
84. Departmental stores selected to test a new merchandising display system is the example of .
a. Quota sampling
b. Convenience sampling
c. Judgmental sampling
d. Purposive sampling
85. In _____, the researcher attempts to control and/ or manipulate the variables in
the study.
1. Experiment
2. Hypothesis
3. Theoretical framework
4. Research design
86. In an experimental research study, the primary goal is to isolate and identify the effect produced by the ____.
1. Dependent variable
2. Extraneous variable
3. Independent variable
4. Confounding variable
87. A measure is reliable if it provides consistent _____.
1. Hypothesis
2. Results
3. Procedure
4. Sensitivity
88. The interview in which questions are already prepared is called _____.
1. Telephonic interview
2. Personal interview

3. Unstructured interview
4. Structured interview
89. The numerical description that describe sample may be expected to differ from those that describe population because of random fluctuations inherent in sampling process.
1. Sampling design
2. Non-probability sampling
3. Sampling error
4. Probability sampling
90. In _____ , each population element has a known and equal chance of selection.
1. Purposive sampling
2. Quota sampling
3. Stratified sampling
4. Simple random sampling
91. _____ is the evidence that the instrument, techniques, or process used to measure concept does indeed measure the intended concepts.
1. Reliability
2. Replicability
3. Scaling
4. Validity
92. A researcher is interested in studying why the "new math" of the 1960s failed. She interviews several teachers who used the new math during the 1960s. These teachers are considered as:
1. Primary sources
2. Secondary Sources
3. External critics
4. Internal critics
93. Which of the following is NOT true about stratified random sampling?
1. It involves a random selection process from identified subgroups
2. Proportions of groups in the sample must always match their population proportions
94. Disproportional stratified random sampling is especially helpful for getting large enough subgroup samples when subgroup comparisons are to be done
4. Proportional stratified random sampling yields a representative sample
95. Experimental design is the only appropriate design where_____ relationship can be established.
1. Strong
2. Linear
3. Weak
4. Cause and Effect
96. Rationalism is the application of which of the following?
A. Logic and arguments
B. Research solution
C. Reasoning
D. Previous findings
97 On which of the following, scientific knowledge mostly relies?

A. Logical understanding
B. Identification of events
C. Prior knowledge
D. All of the given options

98. Which of the following refers to research supported by measurable evidence?

A. Opinion
B. Empiricism
C. Speculation
D. Rationalism

99. Research method is applicable in all of the following fields, EXCEPT;

A. Health care
B. Religion
C. Business
D. Government offices

100. All of the following are true statements about action research, EXCEPT;

A. Data are systematically analyzed
B. Data are collected systematically
C. Results are generalizable
D. Results are used to improve practice

101. Which of the following is characteristic of action research?

A. Variables are tightly controlled
B. Results are generalizable
C. Data are usually qualitative
D. Results demonstrate cause-and-effect relationships

102. If a researcher is studying the effect of using laptops in his classroom to ascertain their merit and worth, he is likely conducting which of the following types of research?

A. Experimental
B. Applied
C. Basic
D. Evaluation

103. Exploratory research addresses which of the following types of question?

A. If
B. How
C. Why
D. What

104. Which of the following is not the source for getting information for exploratory research?

A. Content analysis
B. Survey
C. Case study
D. Pilot study

105. Which of the following is the main quality of a good theory?

A. A theory that has survived attempts at falsification

B. A theory that is proven to be right
C. A theory that has been disproved
D. A theory that has been falsified

106. Which of the following is not a concept?

A. Leadership
B. Total Quality Management
C. Intelligence Quotient (IQ)
D. Human Resource Management

107. A variable that is presumed to cause a change in another variable is known as:

A. Discontinuous variable
B. Dependent variable
C. Independent variable
D. Intervening variable

108. Which of the following is the opposite of a variable?

A. An extraneous variable
B. A dependent variable
C. A data set
D. A constant

109. Which of the following can best be described as a categorical variable?

A. Age
B. Annual income
C. Grade point average
D. Religion

110. "Income distribution of employees" in a specific organization is an example of which of following type of variable?

A. Discontinuous variable
B. Continuous variable
C. Dependent variable
D. Independent variable

111. "There is no relationship between higher motivation level and higher efficiency" is an example of which type of hypothesis?

A. Alternative
B. Null
C. Co relational
D. Research

112. Which of the following is not a role of hypothesis?

A. Guides the direction of the study
B. Determine feasibility of conducting the study
C. Identifies relevant and irrelevant facts
D. Provides framework for organizing the conclusions

113. Hypothesis test may also be called as:

A. Informal test
B. Significance test
C. Moderating test

D. T-test

114. Which type of review compares how different theories address an issue?

A. Context review

B. Integrated review

C. Theoretical review

D. Methodological review

115. After you locate a source, you should write down all details of the reference, EXCEPT;

A. Volumes

B. Titles

C. Price

D. Full names of the authors

For Teacher's use only											
Question Marks	1	2	3	4	5	6	7	8	9	10	Total

Question No: 1 (Marks: 1) - Please choose one

Eye color of students is the example of:

- ► Attribute
- ► Discrete variable
- ► Continuous variable
- ► None of these

Question No: 2 (Marks: 1) - Please choose one

If any value in data is zero, then it is not possible to have:

- ► A.M
- ► Median
- ► Mode
- ► H.M

Question No: 3 (Marks: 1) - Please choose one

The standard deviation of c (constant) is:

- ► C
- ► C^2

► Zero

► None of these

Question No: 4 (Marks: 1) - Please choose one

Scatter diagram is expressed as:

► Curve

► Lines

► Dots

► Rectangles

Question No: 5 (Marks: 1) - Please choose one

The random variable “the number of heads” in tossing of two coins is :

► Continuous variable

► Discrete random variable

► May be discrete or continuous random variable

► None of these

Question No: 6 (Marks: 2)

Define variable and constant.

Question No: 7 (Marks: 2)

Define mutually exclusive events.

Question No: 8 (Marks: 2)

What is the mode of the letters in the Word “ALI”.

Question No: 9 (Marks: 9)

Calculate the **geometric mean** for the distribution given below:

Variable	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	2	5	7	13	21	16	8	3

Question No: 10 (Marks: 10)

Compute and interpret the **correlation co-efficient** for the following data:

X (height)	12	10	14	11	12	9
Y (weight)	18	17	23	19	20	15

For Teacher's use only											
Question Marks	1	2	3	4	5	6	7	8	9	10	Total
Question Marks	11	12	13	14							

Question No: 1 (Marks: 4)

Statistics as a subject, in which two of parts is divided? Expalin briefly both of parts.

Question No: 2 (Marks: 4)

Differentiate simple and composite hypothesis.

Question No: 3 (Marks: 4)

Correct the followings:

$\propto \pm \int$ contains approximately 50% area.

$\propto \pm 2\int$ contains approximately 90% area.

$\propto \pm 3\int$ contains approximately 90.88% area.

Question No: 4 (Marks: 1) - Please choose one

The heights in centimeters of 5 students are:
165, 175, 176, 159, 170.
The sample median and sample mean are respectively:

- 170, 169
- 170, 170
- 169, 170
- 176, 169

Question No: 5 (Marks: 1) - Please choose one

The characteristic which can not be measured numerically is called:

► Quantitative variable

► Qualitative variable

► Discrete variable

► Continuous variable

Question No: 6 (Marks: 1) - Please choose one

The expected value of the normal distribution is

► 0

► 1

► ∞

► $\int$

Question No: 7 (Marks: 1) - Please choose one

Normal distribution is

► Uni-model

► Bi-modal

► Multi-model

► None of these

Question No: 8 (Marks: 1) - Please choose one

One sided and two sided critical regions are based on:

- Level of significance
- Sample size
- Null hypothesis
- Alternative hypothesis

Question No: 9 (Marks: 1) - Please choose one

The rule or formula that is used to estimate a population parameter is called:

- Estimate
- Estimator
- Denominator
- None of these

Question No: 10 (Marks: 1) - Please choose one

The probability of rejecting a true null hypothesis is called:

- Level of significance
- Type-1 error

► Type-II error

► None of above

Question No: 11 (Marks: 1) - Please choose one

The value of chi-square can never be

► Zero

► Negative

► Greater than 1

► None of these

Question No: 12 (Marks: 10)

The grade-point averages of college seniors selected at random from the graduating class are as foll o

3.2	1.9	2.7	2.4
2.8	2.9	3.8	3.0
2.5	3.3	1.8	2.5
3.7	2.8	2.0	3.2
2.3	2.1	2.5	1.0

Calculate the standard deviation.

Question No: 13 (Marks: 10)

The mean lifetime of electric light bulbs produced by a company has in the past been 1120 hours with a standard deviation of 125 hours. A sample of 8 electric bulbs recently chosen form a supply of newly manufactured bulbs showed a mean lifetime of 1070 hours. Test the hypothesis that mean lifetime of the bulbs has not changed using a level of significance of 0.05.

Question No: 14 (Marks: 10)

A random sample of 200 voters is selected and 120 are found to support an annexation suit. Find the 96% confidence interval for the fraction of the voting population favoring the suit.

For Teacher's use only											
Question Marks	1	2	3	4	5	6	7	8	9		Total

Question No: 1 (Marks: 1) - Please choose one

The least squares regression line always goes through the:

- ► Centre of the distribution
- ► -0.5 and +0.5
- ► Means of X and Y
- ► None of the these

Question No: 2 (Marks: 1) - Please choose one

Cumulative frequency polygon is also known as:

- ► Histogram
- ► Pie chart
- ► Frequency Polygon
- ► Ogive

Question No: 3 (Marks: 1) - Please choose one

If any value in the series is negative, the geometric mean will be:

- ► Negative
- ► Positive

► Zero

► Undefined

Question No: 4 (Marks: 1) - Please choose one

When three dice are rolled then number of possible sample points is:

► 6

► 18

► 36

► 216

Question No: 5 (Marks: 1) - Please choose one

Correct formula of m_3 is :

► $3m_{3/} \square 2m_{2/} m_{1/} + \left(m_{1/} \right)_3$

► $m_3 \square 3m_2 m_1 + \left(m_1 \right)_3$

► $m_3 \square 3m_2 m_1 + 2\left(m_1 \right)_3$

► None of the these

Question No: 6 (Marks: 3)

What is five number Summary?

Question No: 7 (Marks: 3)

When $n = 5, \sum X = 30, S_x = 3.286$ **,find** $\sum X^2$ **from the given information.**

Question No: 8 (Marks: 10)

During the first 10 weeks of a session, marks obtained by two students A and B were as follows.

A	58	59	60	54	65	66	52	75	69	52
B	56	87	80	78	72	73	84	65	66	48

Which of the two students A and B was more consistent?

Question No: 9 (Marks: 9)

Show that in a single throw of two dice, the probability of throwing more than 7 is equal to that of throwing less than 7, and hence find the probability of exactly 7.

STA301 Statistics and Probability – I

Mid Term Examination – Spring 2005

Time Allowed: 90 Minutes

Please read the following instructions carefully before attempting any of the questions:

1. **Attempt all questions. Marks are written adjacent to each question.**
2. **Do not ask any question about the contents of this examination from anyone.**
   a. **If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.**
   b. **If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.**
   c. **Write all steps, missing steps may lead to deduction of marks.**
3. **You are allowed to use the calculator & Statistical tables in order to solve the questions.**
4. **For your convenience we are providing you the following symbols,**

□ **,?,** $\bar{X}$ **o**r write Mean, **s,** ∫ or **sd** for standard deviation, s^2 ∫^{2} or sd^2 or variance for variance, √ , □$^{\log x}$,for square root or whole square root.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.**

Solution No. 1 **Marks: 3**

Why is the dispersion of the distribution such as an important characteristic to understand and measure?

Solution No. 2 **Marks: 1**

The marks obtained by 9 students are given below:
45, 32, 37, 46, 39, 36, 41, 48, 36
The arithmetic mean of the values is:

- 39
- 45
- 40
- None of the above

Solution No. 3 **Marks:07**

Deviations from X= 25.5 of 15 different values are given below:

-15.4, -1.9, 6.2, 13.7, 24.6, 25.5, 3.8, -7.9, -13.6, 21.1, 16.4, 18.7, -14.3, -9.8 and 4.9

Calculate the Geometric Mean.

Solution No. 4 **Marks: 1**

In regression problem, the independent variable is:

- Fixed
- Random
- Both of above
- None of above

Solution No. 5 **Marks: 1**

Which of the following is the example of relative measure of dispersion?

- Standard deviation
- Variance
- Coefficient of variation
- All of these

Solution No. 6 **Marks: 03**

What are the two approaches to define probability? Define briefly.

Solution No. 7 **Marks: 1**

If y=x+4 and variance of x is 9 then variance of y is :

- 13
- 9
- 5
- 25

Solution No. 8 **Marks: 1**

Which of the following statements is not correct?

- Some data sets do not have means
- Calculation of a mean is affected by extreme data values
- A weighted mean should be used when it is necessary to take the importance of each value into account
- All of these statements are correct

Solution No. 9 **Marks: 03**

Differentiate between Skewness and kurtosis.

Solution No. 10 **Marks: 01**

For discrete probability distribution, the sum of the probabilities associated with all the possible outcomes must be:

- Zero
- between 0 and 1
- unity
- between -1 and 1

Solution No. 11 **Marks: 01**

As the numbers of observations and classes increase, the shape of a frequency polygon

- Tends to become increasingly smooth
- Tends to become jagged
- Stays the same
- Varies only if data become more reliable

Solution No. 12 **Marks: 07**

A class contains 20 men and 40 women of which half the men and half the women have brown eyes .Find the Probability that a person chosen at random is a man or someone who have brown eyes.